



Synthesis and characterization of Eu-doped $(\text{K}_{0.5}\text{Na}_{0.5})\text{NbO}_3$ ceramics with enhanced ferroelectric, photochromic, and luminescent properties

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ABSTRACT

The presence of secondary phases in rare-earth ion-doped potassium sodium niobate $((\text{K}_{0.5}\text{Na}_{0.5})\text{NbO}_3, \text{KNN})$ ceramics limit its electronic applications. To address this limitation, we investigated the electronic and luminescence properties of Eu³⁺-doped KNN (KNEN) ceramics. We synthesized a high-purity perovskite phase of KNEN without secondary phases by carefully adjusting the Na and K concentrations with 1 mol% Eu. KNEN ceramics demonstrated excellent ferroelectric and luminescent properties, including a photochromic effect that presented reversible color changes on exposure to ultraviolet light and X-rays. Further, we observed a strong correlation between the electric and luminescent properties near the orthorhombic-tetragonal phase transition at $\sim 200^\circ\text{C}$. These results suggest that KNEN ceramics are promising Pb-free luminescent piezoelectric materials.

1. Introduction

Lead-based piezoelectric ceramics, such as $\text{Pb}(\text{Zr,Ti})\text{O}_3$ (PZT), exhibit excellent ferroelectric and piezoelectric properties, making them suitable for applications in ultrasound, sensors, and actuators [1]. However, these lead-based materials pose significant risks to human health and the environment owing to the toxicity of lead [2]. Therefore, the Pb-free piezoelectric ceramics have been extensively studied as potential replacements for achieving comparable piezoelectric and electronic properties [3–5].

Potassium sodium niobate $(\text{K}_{0.5}\text{Na}_{0.5}\text{NbO}_3, \text{KNN})$ is Pb-free piezoelectric ceramics that can potentially replace PZT ceramics. KNN has attracted significant research attention because of its excellent piezoelectric constant and high Curie temperature, comparable to those of Pb-containing ferroelectrics. The outstanding properties of KNN make it suitable for applications in actuators and sensors [6], energy harvesting [7–10], and memory devices [11]. When KNN is synthesized using the conventional solid-state reaction method, K and Na ions can volatilize at very high temperatures [12]. Co-doping cations with rare-earth (RE) ions is an effective approach for suppressing the cation deficiency and increasing the piezoelectric properties. Therefore, understanding the simple RE doping effect of KNN ceramics is necessary to applying them to modified KNN systems. Rutkowski et al. produced highly dense

ceramics via doping with RE ions such as Er^{3+} and Yb^{3+} , and modifying the fabrication process [13]. The Eu³⁺-doped $\text{K}_{1-x}\text{Na}_x\text{NbO}_3$ ($x = 0.42\text{--}0.56$) ceramic showed the highest piezoelectric property at $x = 0.50$ [14]. The highest Curie temperature was observed for a 1 mol% Eu-doped KNN ceramic [15]. A small amount of the impurity phase can be observed in the X-ray diffraction (XRD) patterns because of the volatilization of the K and Na during the high-temperature heat-treatment process. The melting point of KNbO_3 ($T_m = 1050^\circ\text{C}$) is lower than that of NaNbO_3 ($T_m = 1422^\circ\text{C}$) [16]. In addition, RE-doped KNN ceramics require a 50–100 °C higher sintering temperature than that of undoped KNN ceramics to achieve high densification. This increased sintering temperature exceeds the melting point of the KNN ceramics, which is $\sim 1140^\circ\text{C}$ [17,18]. Therefore, controlling the formation of secondary phases by adjusting the K and Na contents of the starting composition is essential. Further, this approach can demonstrate the effects of Eu doping without the interference from secondary phases.

The KNN ceramic exhibits a phase transition from the orthorhombic to the tetragonal ($T_{\text{O-T}}$) phase at 200°C and from the tetragonal to cubic ($T_{\text{T-C}}$) phase at 420°C [19,20]. The piezoelectric performance is greatly improved when the $T_{\text{O-T}}$ is reduced to room temperature to dope transition metals (such as Ta and Sb) or add other perovskite compounds [21]. However, this decreases $T_{\text{T-C}}$ (Curie temperature). In addition, multi-ion doping or a solid solution with another ABO_3 is required to

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improve the temperature stability of piezoelectric properties. This process can result in inhomogeneities between the grains, causing variations in the piezoelectric response at different points within the material.

Recently, RE doped lead-free piezoelectric ceramics have attracted considerable research attention because of their novel optical properties such as transparency, photoluminescence, and photochromism, along with their original electrical properties [22–30]. Among RE ions, trivalent europium (Eu^{3+}) ions are notable for their orange-red emission spectra, which originate from the 4f-4f transitions ($^5\text{D}_0\text{--}^7\text{F}_J$, $J = 0\text{--}5$). The crystal structure of the host material affects the emission characteristics of Eu^{3+} because of the high sensitivity of the 4f-4f transition to the local lattice structure [31–34]. KNN is a promising photoluminescent host because its low phonon energy and high thermal resistance [35]. $(\text{K}_{0.5}\text{Na}_{0.5})_{1-x}\text{Eu}_x\text{NbO}_3$ ($x = 0.02\text{--}0.30$) ceramics exhibit photochromism and energy storage behavior with a secondary phase (EuNbO_4) [36]. The $x\text{Eu}$ -KNN ($x = 1\text{--}4\%$) exhibits optical transmittance and photochromic (PC) behavior [15].

In this study, $(\text{K}_{0.50}\text{Na}_{0.50})_x\text{Eu}_{0.01}\text{NbO}_3$ (KNEN) ceramics with $x = 0.99, 1.00$, and 1.01 were synthesized using the conventional solid-state reaction method. The second phase was controlled by adjusting the K and Na content in the starting composition. The ferroelectric and piezoelectric properties were found to depend on the x content. In addition, photochromism and related photoluminescence (PL) modulation under ultraviolet (UV) irradiation were observed, which were attributed to the effect of Eu^{3+} doping.

2. Experimental

2.1. Ceramic preparation

Eu^{3+} -doped KNN (KNEN) ceramics were prepared using a conventional solid-state reaction method. The raw materials used were K_2CO_3 (99 %, Aldrich), Na_2CO_3 (99 %, Aldrich), Nb_2O_5 (99.9 %, Aldrich), and Eu_2O_3 (99.9 %, Aldrich). Before weighing, K_2CO_3 and Na_2CO_3 powders were dried at 150°C for 1 h. The powders were then weighed according to the compositions $(\text{K}_{0.5}\text{Na}_{0.5})_x\text{Eu}_{0.01}\text{NbO}_3$ ($x = 0.99, 1.00$, and 1.01). The details regarding the weight of powders are presented in Table S1. The weighed powders were milled with zirconia balls in ethanol for 24 hours, then dried at 100°C , and calcined at 850°C for 4 h. After calcination, polyvinyl alcohol (PVA) was added to the powder as a binder. The green ceramics were prepared by uniaxial pressing into pellets with a thickness of 1 mm and a diameter of 10 mm. The green ceramics were covered with the same powder prior to sintering to prevent the volatilization of K and Na. Subsequently, ceramics were heated to 850°C for 1 h to effectively burn out the PVA binder, followed by sintering at 1170°C for 3 h. The photograph of before and after sintering state is provided in Figure S1.

2.2. Measurement method

The crystal structures were determined using the XRD (AXS D2 system, Bruker) and the surface morphologies were observed using a scanning electron microscope (SEM, JSM-6510, JEOL). The electrical properties of the KNEN ceramics were measured by applying the silver paste to a polished surface (ceramic thickness: $600\text{ }\mu\text{m}$) followed by baking at 350°C for 30 min. The ferroelectric polarization–electric field (P - E) hysteresis loops were measured using a ferroelectric measurement system at 1 Hz (RT6000 HVS, Radiant). The ceramics were poled using a 30 kV/cm DC electric field for 30 min at room temperature. After poling, d_{33} was measured using a piezo- d_{33} -meter with 0.25 N and 110 Hz (ZJ-6B, IACAS). The temperature-dependent dielectric constant ($\epsilon_r(T)$), dielectric loss ($\tan \delta(T)$), and resonance properties were measured using an impedance analyzer (HP4194A, Agilent). The planar piezoelectric coupling factor, k_p was evaluated using the resonance anti-resonance method [European Standard (EN 50324–2)] according to,

$$k_p = \sqrt{(0.395 \frac{f_r}{f_a - f_r} + 0.574)^{-1}} \quad (1)$$

where f_r and f_a represent the resonance and anti-resonance frequencies, respectively.

The PL and PL excitation (PLE) spectra were recorded using a fluorometer (FP-8500, JASCO) [37]. The diffuse reflectance spectra of the ceramics were measured in the near-infrared-visible-UV range ($0.5\text{--}6.7\text{ eV}$) using a UV spectrophotometer (V-670, JASCO). The UV light at 380 nm and X-ray ($\lambda = 1.5279\text{ \AA}$) were irradiated on the samples to detect a PC behavior.

3. Results and discussion

3.1. Structural analysis on KNEN

Fig. 1(a) shows the XRD patterns of KNEN ceramics with varying K and Na contents ($x = 0.99, 1.00$, and 1.01). The XRD pattern for the ceramic with $x = 1.00$ indicates a pure perovskite structure without any secondary phases. In contrast, XRD patterns for ceramics with $x = 0.99$ and 1.01 reveal a perovskite structure along with a secondary phase, with peaks marked by (peak position at 25° to 30°). This secondary phase corresponds to a tungsten bronze-type structure of $\text{K}_6\text{Nb}_{10.9}\text{O}_{30}$, which is associated with an A-site deficient state of KNbO_3 [38,39].

The secondary phase of the $x = 0.99$ ceramic is already present in the calcined powder, as shown in Fig. S2. This indicates a K and Na deficiency in the calcined powder state of the $x = 0.99$ ceramic. In contrast, the $x = 1.00$ ceramic exhibits a single perovskite structure in both the calcined powder and sintered ceramic, indicating a stoichiometric composition throughout the process. For the $x = 1.01$ ceramic, the secondary phase was only detected in the sintered ceramic, because of the residual K and/or Na reacting during the sintering process [40–42].

Interestingly, the lattice constants of KNEN changed with x . Fig. 1(b) shows the (200) and (020) peaks from 44.5 to 46.5° . The double peaks are typical reflection patterns of orthorhombic phase of KNN ceramic with $Amm2$ space group. These distinct splitting patterns have been observed from other literature [43,44]. Although the peak centers are nearly identical for $x = 0.99$ and $x = 1.00$, the peak centers of the (200) and (020) planes of the $x = 1.01$ ceramic are shifted to a higher angle from the peak center of (200) and (020) $x = 1.00$. This implies that the lattice constant of the $x = 1.01$ ceramic is also considerably smaller lattice constant compared with the $x = 1.00$. Currently, the origin of this behavior is not clearly understood.

For the quantitative analysis of the structural properties, we employed the Rietveld refinement of the XRD patterns of the $x = 1.00$ KNEN. The refinement results are shown in Fig. 2. The $x = 1.00$ ceramic is fitted with an orthorhombic phase ($Amm2$) and their reliability factors are $R_{wp} = 16.9$ and $\chi^2 = 1.58$. The lattice parameters are calculated with $a = 3.944\text{ \AA}$, $b = 5.636\text{ \AA}$ and $c = 5.661\text{ \AA}$. The Rietveld refinement analysis for both $x = 0.99$ and $x = 1.01$, and the results are presented in Fig. S3 (a) and (b). For the $x = 0.99$ ceramic, the refinement indicates that the composition consists of 91.8 % KNN and 8.2 % KN. In the case of the $x = 1.01$ ceramic, the analysis shows a composition of 95.4 % KNN and 4.6 % KN. The fitting parameters are listed in Table 1.

We also synthesized powders at $x = 0.98$ and 1.02 by conventional solid solution method, as detailed in the experiment part. Fig. S4. Presents the X-ray diffraction (XRD) patterns of the KNEN powders across various compositions, specifically $x = 0.98, 0.99, 1.00, 1.01$, and 1.02 . Notably, the $x = 0.98$ and 0.99 compositions exhibited distinct peaks corresponding to the secondary phase $\text{K}_6\text{Nb}_{10.9}\text{O}_{30}$. In contrast, the $x = 1.00, 1.01$, and 1.02 compositions did not display evidence of secondary phases. The XRD results for the sintered ceramics are illustrated in Figure S5. The samples with $x = 0.98$ and 0.99 again show the presence of the secondary phase, while the $x = 1.00$ composition solely exhibits an orthorhombic phase. Additionally, the $x = 1.01$ and 1.02 samples also

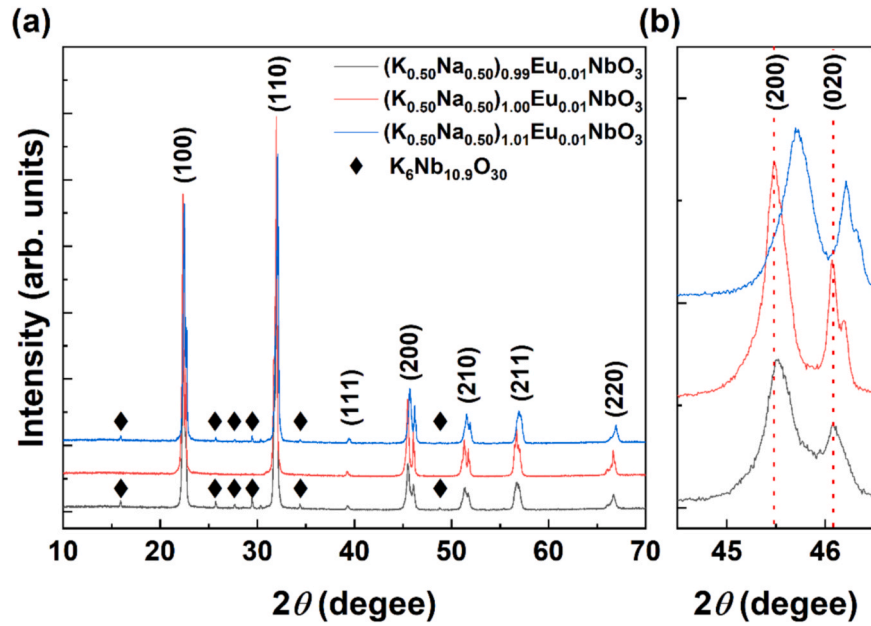


Fig. 1. (a) X-ray diffraction patterns of KNEN for $x = 0.99, 1.00$, and 1.01 in 2θ ranges of (a) $10 - 70^\circ$ and (b) $44.5 - 46.5^\circ$. The solid diamond symbols represent XRD peaks from the secondary phase K deficient $K_6Nb_{10.9}O_{30}$.

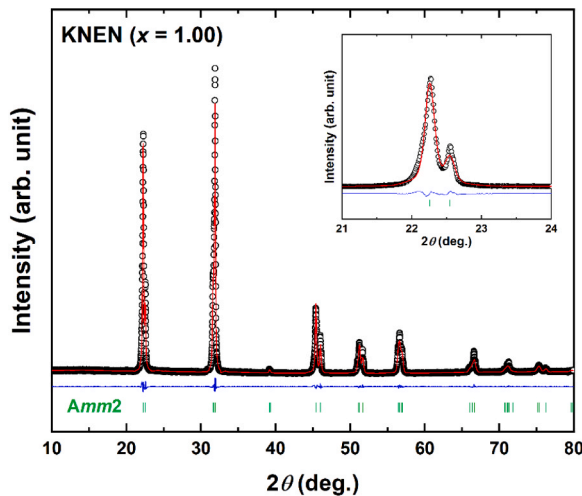


Fig. 2. Refinement result of the X-ray diffraction patterns of the $x = 1.00$ KNEN. The solid thick lines (red), open circles, solid thin lines (blue), and green bars represent the calculated data, experimental data, subtracted data (that is, $\text{Exp.} - \text{Cal.}$), and Bragg positions, respectively. Inset: Magnified view of (100) peak.

Table 1

Results of the Rietveld refinement analysis on the XRD pattern of the $x = 0.99, 1.00$, and 1.01 KNEN.

$(K_{0.5}Na_{0.5})_xEu_{0.01}NbO_3$					
x	0.99		1.00	1.01	
Space group	Am2	P4/mbm	Am2	Am2	P4/mbm
Phase	Orthorhombic	Tetragonal	Orthorhombic	Orthorhombic	Tetragonal
a(Å)	3.94802	12.5393	3.94371	3.93713	12.54628
b(Å)	5.63671	12.5393	5.635931	5.61474	12.54628
c(Å)	5.66093	3.96870	5.66090	5.64250	3.96419
$\alpha(^{\circ})$	90	90	90.00	90	90
$\beta(^{\circ})$	90	90	90.00	90	90
$\gamma(^{\circ})$	90	90	90.00	90	90
V (Å ³)	125.977	624.015	125.822	124.733	624.011
χ^2	3.38		1.58	3.61	
R _{wp} (%)	16.8		16.9	16.4	

display secondary phase peaks.

3.2. SEM images

The SEM images of KNEN are shown in Fig. 3(a)–(c). The physical properties of the ceramics, including the crystal structure and the ferroelectric, piezoelectric, dielectric, and optical properties, are significantly affected by the grain distribution and ceramic density. Fig. 3(a) – (c), show that the number of pores decreased with an increase in the Na and K contents from $x = 0.99$ to $x = 1.00$. Conversely, the number of pores increased from $x = 1.00$ to $x = 1.01$. Uniform grain sizes were observed in $x = 1.00$ (Fig. 3(b)). The grain size of the KNEN ceramics was analyzed through SEM images in Fig. S7. It was found that most grain sizes across all compositions are below $1 \mu\text{m}$. Specifically, the mean grain sizes were measured as follows: $0.895 \mu\text{m}$ for $x = 0.99$, $0.801 \mu\text{m}$ for $x = 1.00$, and $0.796 \mu\text{m}$ for $x = 1.01$. The ceramic densities of the sintered samples were measured using Archimedes' method. The density of the $x = 0.99$ ceramic was measured at 4.36 g/cm^3 , whereas the density increased to 4.48 g/cm^3 for $x = 1.00$, achieving a relative density of 97 %. Further, the density decreased to 4.38 for $x = 1.01$ [45]. The high densification of the KNN ceramic is attributed to the addition of Na_2O and K_2O [46]. In the case of $x = 0.98$ (4.15 g/cm^3) and 1.02 (4.21 g/cm^3), the ceramic density is smaller than $x = 0.99$ and 1.01 as shown in Fig. S6. Additionally, the cell volume of the secondary phase

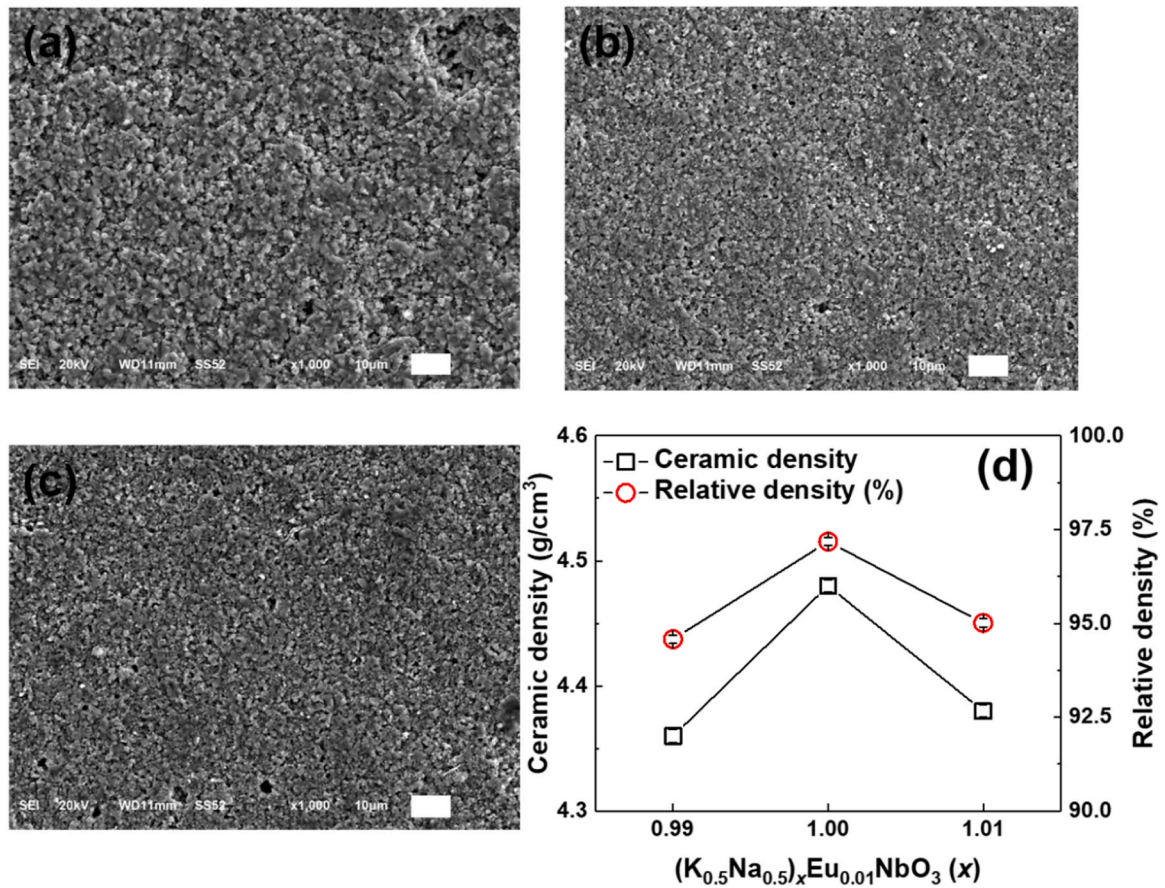


Fig. 3. Scanning electron microscope images of KNEN for (a) $x = 0.99$, (b) $x = 1.00$, and (c) $x = 1.01$. (d) Density (black squares) and relative density (red circles) values of KNEN.

$\text{K}_6\text{Nb}_{10.9}\text{O}_{30}$ is $624 \text{ \AA}^3$, which is considerably larger than that of $(\text{K}_{0.5}\text{Na}_{0.5})\text{NbO}_3$, which has a cell volume of $125 \text{ \AA}^3$. The presence of this secondary phase tends to grow between the grains, potentially leading to pore formation at the grain boundaries. This phenomenon may result in a reduction of the overall relative density of the ceramic, highlighting the intricate balance between phase composition, microstructure, and densification in the processing of KNEN ceramics.

3.3. Electric properties of KNEN: P - E loops and d_{33}

The ferroelectric and piezoelectric properties of KNEN ceramics with $x = 0.99$, 1.00 , and 1.01 are shown in Fig. 4. Ferroelectric hysteresis loops are measured at 1 Hz at room temperature (Fig. 3(a)). The ferroelectric properties are characterized by remnant polarization ($2P_r$), saturated polarization ($2P_s$), and coercive field ($2E_c$). As shown in Fig. 3(a), the highest values of $2P_r$ and $2P_s$ were observed for $x = 1.00$ composition, with $2P_r$ and $2P_s$ values of 41 and $56 \mu\text{C}/\text{cm}^2$, respectively, as summarized in Fig. 3(b). The lowest coercive field ($2E_c$) of $18.7 \text{ kV}/\text{cm}$ is observed $x = 1.00$. The piezoelectric coefficient (d_{33}) shown in Fig. 3(c), was measured 24 h after poling with a DC bias. d_{33} reached its highest value of $135 \text{ pC}/\text{N}$ at $x = 1.00$. Further, the electromechanical coupling coefficient (k_p) and phase angle (θ) were measured for poled ceramics, as shown in Fig. 4(d). The k_p was maximized at $x = 1.00$, and the highest phase maximum (θ_{max}) was observed at this composition.

The ferroelectric and piezoelectric properties of materials depend on the number of free and bound charges. The bound charge is associated with the charge arising from the induced dipole moments, while the net sum of the free and bound charges is referred to as the true charge of the ferroelectric capacitor [47]. During the ferroelectric hysteresis loop measurements, both free and bound charges accumulated within the

ferroelectric layer. However, the presence of a secondary phase can impede the accumulation of true charge, leading to a reduction in both saturated and remnant polarizations. In addition, the secondary phase increases the coercive field, hindering domain switching within the ferroelectric material.

For piezoelectric behavior, when the external electric field is zero and stress is applied to the ceramic, a charge is generated because of the external stress. In this state, the relationship is given by:

$$d_{33} = 2\varepsilon_0\varepsilon_r Q_{33}P_s \quad (2)$$

where ε_0 , ε_r , Q_{33} , and P_s represent the dielectric permittivity of free space, relative permittivity, electrostrictive constants, and saturated polarization, respectively. According to Eq. (2), the piezoelectric coefficient d_{33} is directly proportional to P_s . Therefore, the d_{33} value for $x = 1.00$ is the highest compared to those of the $x = 0.99$ and $x = 1.01$ ceramics. The electromechanical coupling coefficient (k_p) represents the efficiency of the electromechanical transduction and is proportional to the piezoelectric coefficient. Fig. 4(c) shows that k_p reaches its maximum value at $x = 1.00$. The presence of secondary phases reduces polarization during the ferroelectric hysteresis loop, which also affects the impedance phase results. The phase angle (θ_{max}) changes from -90° to 90° ($\theta_{\text{max}} = 90^\circ$) [48,49]. In ceramics with deficient or excess K and Na, the domain alignment under an external field is weaker because of the presence of secondary phases, which leads to a reduced phase at the resonance frequency. The optimal K and Na content in KNEN ceramics results in the highest θ_{max} . Therefore, the observed ferroelectric, piezoelectric, and electromechanical coupling properties are consistent with the XRD results.

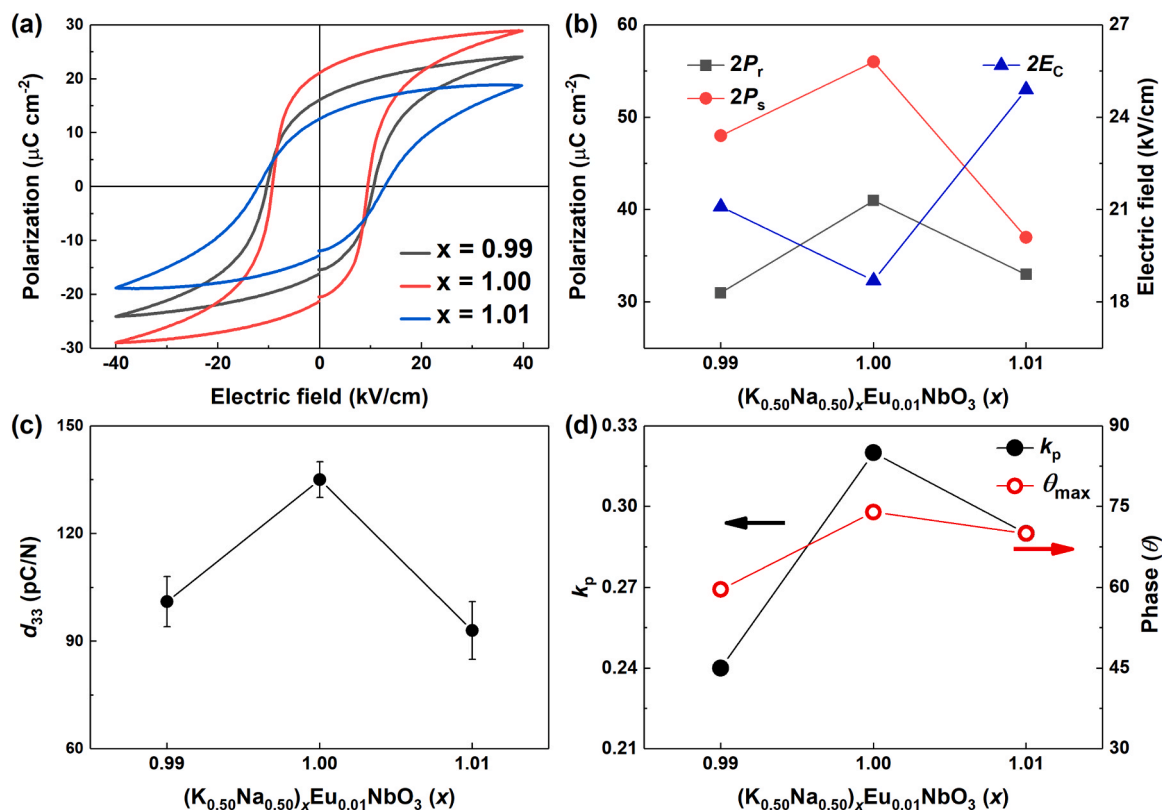


Fig. 4. (a) Ferroelectric hysteresis loops of KNEN for $x = 0.99$, 1.00 , and 1.01 . (b) Remnant (P_r) and saturated polarization (P_s) and coercive field (E_c) of KNEN. (c) Piezoelectric coefficient (d_{33}) of KNEN. (d) Electromechanical coupling (k_p) and phase (θ) for KNEN.

3.4. Photoluminescence properties of KNEN

Fig. 5 shows the PLE and PL spectra of KNEN ceramics with $x = 0.99$, 1.00 , and 1.01 . The PLE spectra were obtained by monitoring the 613 nm emission ($^5\text{D}_0 \rightarrow ^7\text{F}_2$ in Eu^{3+}) [50–52]. Fig. 5(a) shows an intense peak at 465 nm ($^7\text{F}_0 \rightarrow ^5\text{D}_2$ in Eu^{3+}). On excitation at 465 nm, five typical emission bands were observed at 579, 593, 615, 650, and 700 nm, respectively. The peaks were assigned to the transitions of $^5\text{D}_0 \rightarrow ^7\text{F}_J$ ($J = 0, 1, 2, 3$ and 4) in sequence [53]. The highest PLE and PL intensities in KNEN were observed for $x = 1.00$. The photoexcitation volume of the sample is a critical factor in determining the luminescence intensity. For accurate comparisons of PL intensity among samples, it is essential to ensure uniform photoexcitation volumes (or uniform sample

thickness). The photoexcitation volume is primarily influenced by the sample's density. The density variation across our samples is only 2 %, indicating that the photoexcitation volumes are nearly identical for all three samples. However, the observed 60 % variation in PL intensities significantly exceeds the density variation. This suggests that the maximum PL intensity observed at $x = 1.00$ is primarily due to the intrinsic properties of the samples. Additionally, we measured the photoluminescence quantum yield (PLQY) and found that the PLQY is highest for the $x = 1.00$ sample, as shown in Fig. S8. This further confirms that the highest PL intensity at $x = 1.00$ is an intrinsic property of the material. The highest PL signal at $x = 1.00$ implies that optimal K and Na contents and the absence of any secondary phase are important for luminescent as well as the ferroelectric and piezoelectric properties.

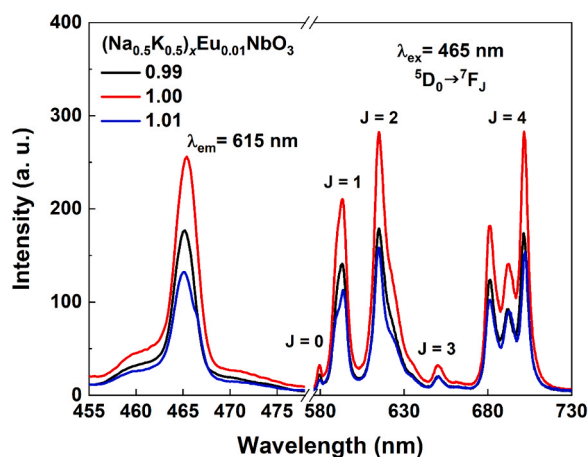


Fig. 5. Photoluminescence excitation and photoluminescence spectra of KNEN for $x = 0.99$, 1.00 , and 1.01 .

3.5. Photochromic behaviors of KNEN

We discussed the PC behavior of KNEN. The PC behavior is characterized by color conversion after irradiation with a short-wavelength light (e.g., UV light), which is typically associated with the generation of electron–hole pairs or a phase transition. The colored states return to their initial states through thermal or visible-light irradiation [54]. In KNEN, electrons excited by UV light are trapped at defect sites, whereas holes are captured by A-site defects, such as those from potassium (K) and sodium (Na) volatilization or europium (Eu) ion transitions. This process leads to the formation of F-colored centers, which are associated with trapped electrons and reduced oxygen vacancies [55]. We note that the oxygen vacancy is important for understanding photochromic behavior. We measured the XPS spectra to confirm the existence of the oxygen vacancy within the samples. The results are shown in Fig. S11. The oxygen peaks are calibrated by the C1s peak, which serves as a reference for each sample. The peaks were deconvoluted using the xpspeak 4.1 program. The peaks were fitted with two modes originating from the lattice oxygen (O_1), and the oxygen vacancies (O_2) [28,29]. We

confirmed that all three samples possess a considerable amount of oxygen vacancies.

Our samples exhibited good PC behavior. As shown in Figs. 6(a)–6(c), there was a notable reduction in the reflectivity across the 400–1000 nm range after UV irradiation at 380 nm for 3 min, which led to the coloration of our samples. ΔR was calculated to quantify the degree of color change using the formula $\Delta R = (R_0 - R_t) \times 100\%$, where R_0 and R_t represent the reflectivity at 500 nm before and after UV irradiation, respectively. The highest ΔR value was observed for the $x = 1.00$ sample, which reached 43 % (Fig. 6(d)). When the colored sample was heated up to 200 °C and annealed for 300 seconds, the reflectivity spectra returned to the original level, and the broad absorption in range of 400–1000 nm disappeared, thereby effectively bleaching the gray coloration (Fig. S9). The photos of the samples in coloring and recovery are shown in Fig. S10 of Supplement.

3.6. PL modulation of KNEN in relation to PC behavior

The PC behavior of the KNEN significantly influenced its PL properties. Fig. 7 shows the PL spectrum of KNEN after the UV light irradiation for 3 min. After UV irradiation, the luminescence intensity decreased, although the shape and peak positions of the PL spectra remained unchanged. The change in luminescence intensity can be quantified using [56]

$$\Delta I_{\text{dec}} = |I_t - I_0|/I_0 \times 100\% \quad (3)$$

where I_0 and I_t represent the initial emission peak intensities at 593, 615 nm and after the UV light irradiation, respectively. The ΔI_{dec} of KNEN were estimated to be 69, 80, and 72 % for $x = 0.99$, 1.00, and 1.01, respectively, indicating that the PL modulation was the largest at $x = 1.00$. Further, we observed that the PL intensity could be restored by a bleaching process involving thermal excitation. When the samples were

heated to 200 °C, the luminescence intensity increased significantly, whereas the shape and peak positions of the PL spectra remained unchanged.

Fig. 8 reveals the five cycles of PL modulation for the KNEN sample under alternating 380 nm UV irradiation and thermal excitation up to 200 °C. The results indicate no significant degradation in ΔI_{dec} and only minor fluctuations, demonstrating the excellent repeatability of the KNEN material. Therefore, our results indicate that KNEN ceramics have potential applications in erasable optical storage devices.

We found that the PC behaviors of the samples can also be induced by X-ray irradiation [57], similar to that with UV irradiation. When exposed to X-rays ($\lambda_{\text{ex}} = 1.54184 \text{ \AA}$), the samples turned gray, indicating the occurrence of PC behavior. Simultaneously, as shown in Fig. 9(b), the PL intensity decreased, which is comparable to the reduction observed under UV irradiation (Fig. 9(a)). Upon heating the samples to 200 °C, the gray coloration was bleached, and the PL intensity returned to its original level. These results indicate that our samples exhibited PC behaviors and related PL modulation under both UV light and X-ray irradiation.

3.7. Structural phase transition and related changes in electric and luminescent properties

A close correlation can be observed between the electrical and luminescent properties in relation to the structural phase transition. We measured the temperature-dependent dielectric permittivity (ϵ) and dielectric loss ($\tan \delta$) of KNEN for $x = 1.00$ up to $T = 220 \text{ °C}$. As shown in Fig. 10(a), the dielectric permittivity of KNEN exhibited a broad peak near $T_m = 180 - 200 \text{ °C}$, which corresponds to the transition from the orthorhombic to the tetragonal phase. This broad peak suggests that the samples behaved as relaxor ferroelectrics rather than normal ferroelectrics with a sharp phase transition [32].

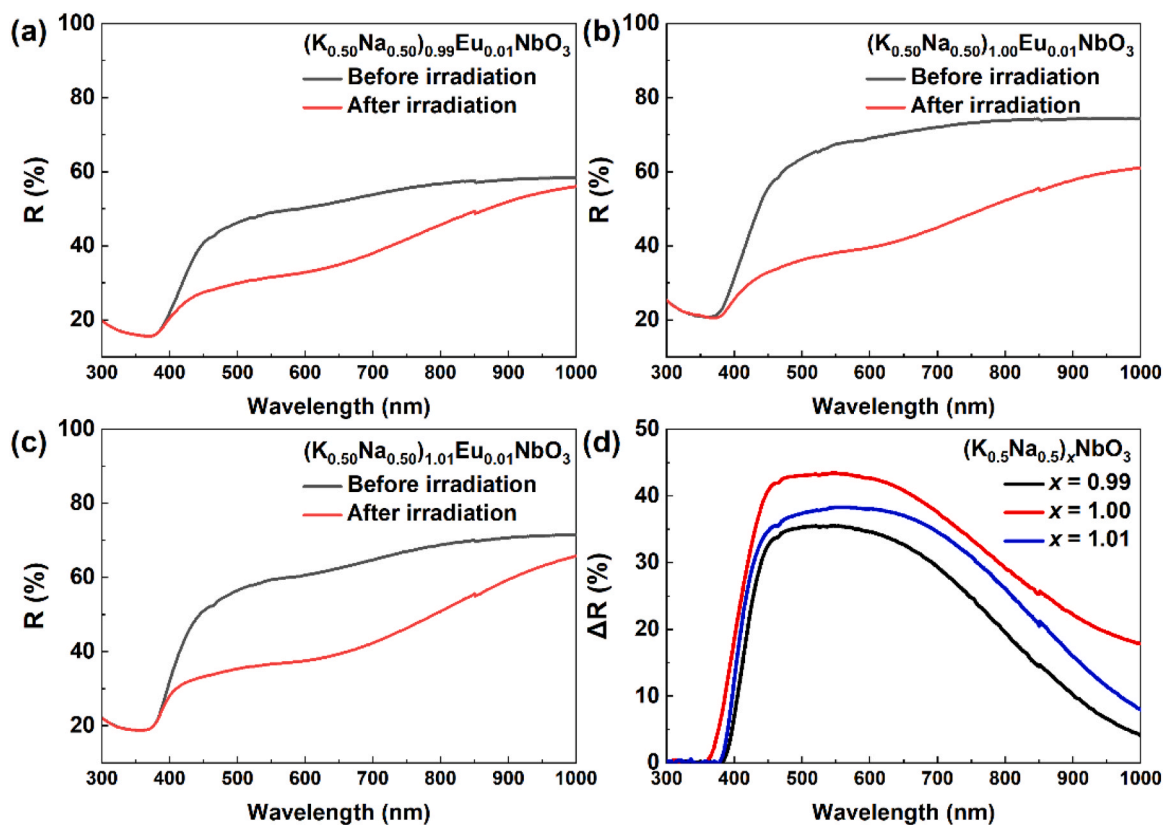


Fig. 6. Reflectivity spectra of KNEN before and after UV ($\lambda = 380 \text{ nm}$, 3 min) irradiation for (a) $x = 0.99$, (b) $x = 1.00$, and (c) $x = 1.01$. (d) Substrate reflectivity spectra ($R_{\text{before}} - R_{\text{after}}$) of KNEN.

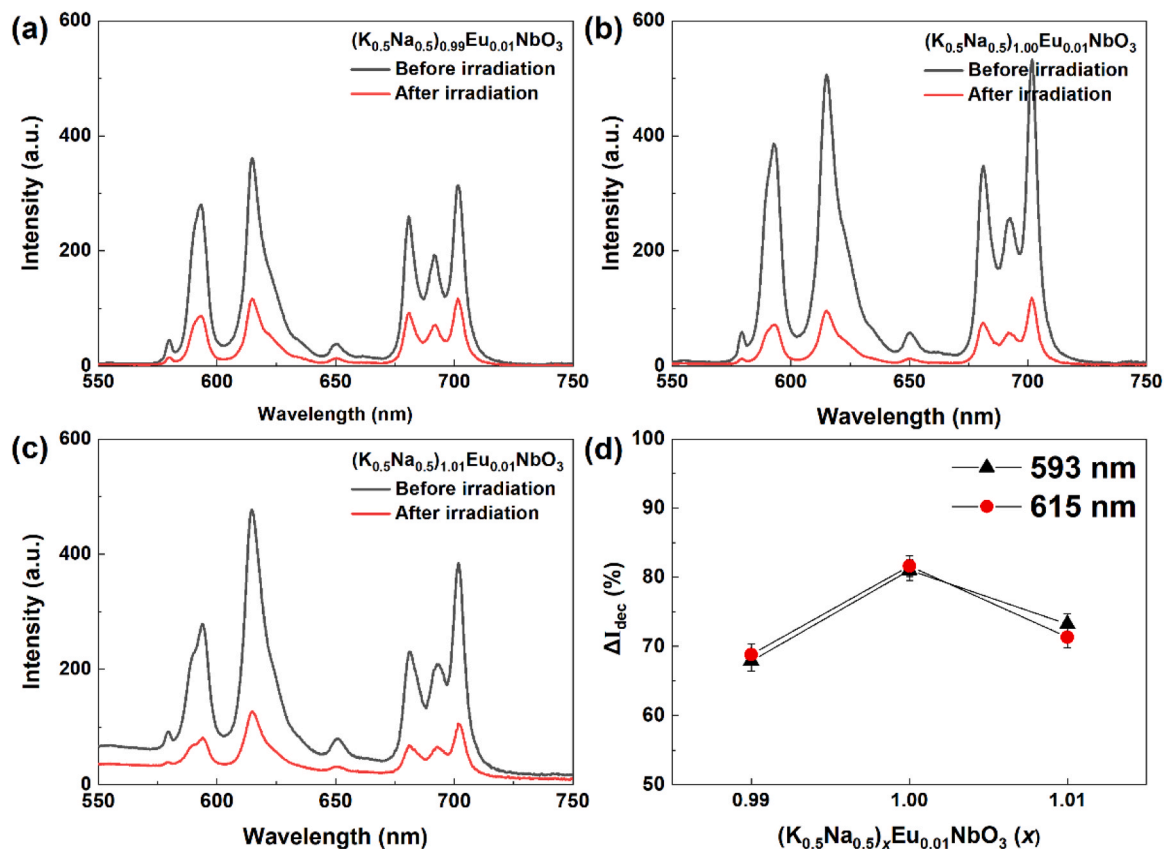


Fig. 7. PL spectra of KNEN before and after UV ($\lambda = 380$ nm) irradiation for (a) $x = 0.99$, (b) $x = 1.00$, and (c) $x = 1.01$. (d) ΔI at 593 and 615 nm for KNEN as a function of x .

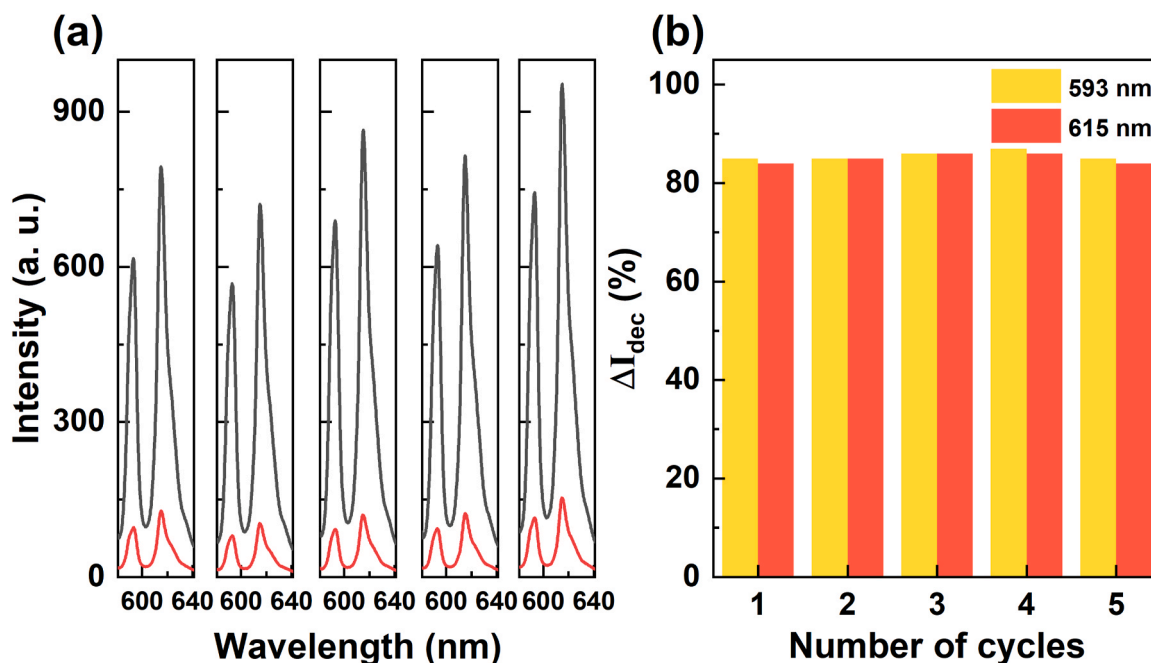


Fig. 8. (a) PL spectra of the $x = 1.00$ KNEN via alternating UV irradiation ($\lambda = 380$ nm, 3 min) and thermal treatment (300°C , 3 min). (b) ΔI_{dec} at 593 and 615 nm during five cycles for the $x = 1.00$ KNEN.

The T-dependent PL spectra of the $x = 0.01$ sample were measured (Fig. S12). The PL peaks were suppressed with increasing temperature. For further insight, we estimated the ratio of the PL intensities at 595

and 615 nm, which corresponds to an asymmetric ratio [58]. As shown in Fig. 10(b), the asymmetrical ratio changed from negative to positive near T_m , indicating a close connection with the structural phase

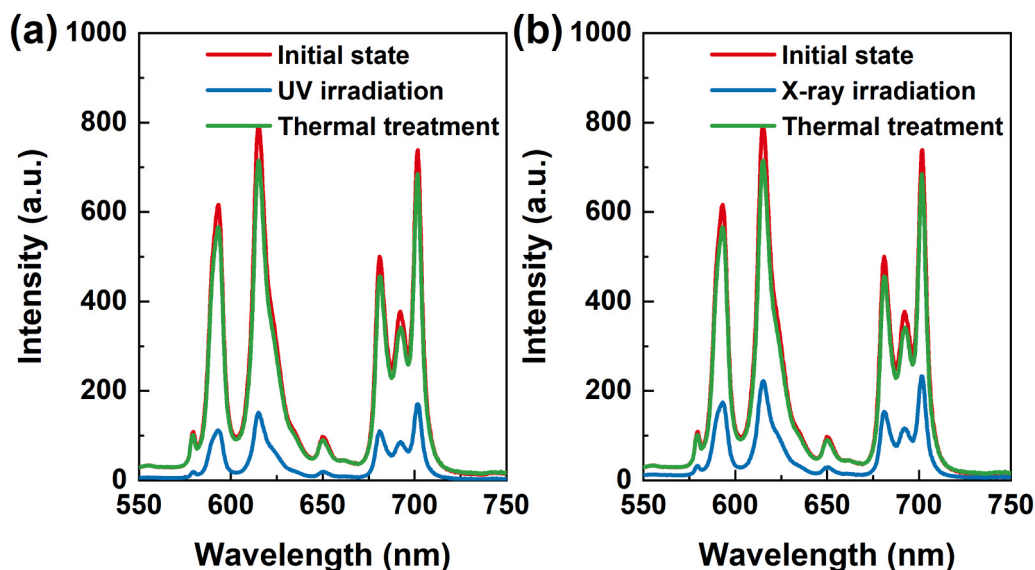


Fig. 9. Change in photoluminescence spectra under (a) UV and (b) X-ray irradiation for the $x = 1$ KNEN.

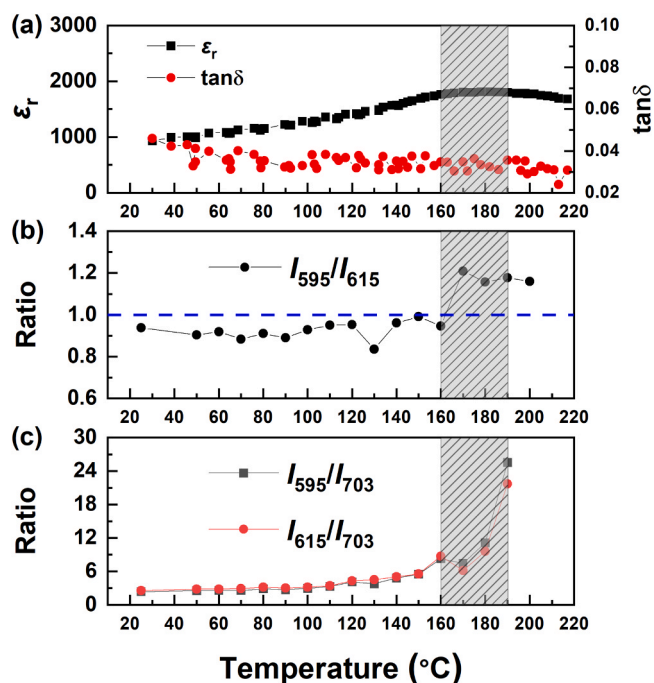


Fig. 10. (a) Temperature-dependent dielectric constant and dielectric loss. (b) Asymmetry ratio (I_{595} versus I_{615}) and (c) I_{595}/I_{703} and I_{615}/I_{703} as a function of temperature.

transition. Further, we examined the relative PL intensities at 595 and 615 nm with respect to that at 703 nm (Fig. 10(c)). These relative intensities show drastic changes near T_m . These findings indicate that the PL spectra of KNEN reflect the underlying structural phase transitions.

4. Concluding remarks

The structural and luminescence properties of the Eu^{3+} doped KNN (KNEN) ferroelectric compounds were investigated in this study. We successfully synthesized a high purity of perovskite phase of KNEN without any secondary phase by fine-tuning the Na/K concentration ratio to 1 mol% Eu. The KNEN ceramics demonstrate excellent ferroelectric and luminescent properties, including a notable photochromic

effect. The dual photochromic effect induced by X-ray and UV irradiation highlighting their potential as promising luminescent ferroelectric compounds. The phase transition near 200 °C between the orthorhombic to the tetragonal phase defined by temperature-dependent dielectric measurement was identified in the temperature-dependent PL spectra. These findings indicate that secondary-phase-free KNEN ceramics are excellent candidates for multifunctional lead-free ferroelectric materials with multifunctional applications.

CRediT authorship contribution statement

Yun Sang Lee: Writing – review & editing, Writing – original draft, Supervision, Resources, Project administration, Funding acquisition, Data curation. **Tae Kwon Song:** Supervision, Resources, Funding acquisition, Data curation. **Myang Hwan Lee:** Investigation, Formal analysis, Data curation. **Jae Ho Han:** Methodology, Investigation, Formal analysis. **Hae In Choi:** Writing – review & editing, Writing – original draft, Methodology, Investigation, Formal analysis, Data curation, Conceptualization. **Chae Yeon Lee:** Data curation, Formal analysis, Investigation, Resources.

Declaration of Competing Interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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Appendix A. Supporting information

Supplementary data associated with this article can be found in the online version at [doi:10.1016/j.jallcom.2024.177690](https://doi.org/10.1016/j.jallcom.2024.177690).

Data availability

No data was used for the research described in the article.

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